

Problem 15

Based on the general term of the series, a_n , conclude that the following series diverges:

$$\sum_{n=0}^{\infty} \cos(n\alpha) \quad \left(-\frac{\pi}{2} < \alpha < \frac{\pi}{2}\right).$$

Solution

Step 1: Identify the General Term

The general term of our series is:

$$a_n = \cos(n\alpha), \quad \text{where } -\frac{\pi}{2} < \alpha < \frac{\pi}{2}.$$

Step 2: Analyze the Behavior of the General Term

For the series to converge, a necessary condition is that:

$$\lim_{n \rightarrow \infty} a_n = 0.$$

We need to determine whether $\lim_{n \rightarrow \infty} \cos(n\alpha)$ exists and equals zero.

Step 3: Consider Different Cases

Case 1: $\alpha = 0$

If $\alpha = 0$, then:

$$a_n = \cos(0) = 1 \quad \text{for all } n.$$

Thus:

$$\lim_{n \rightarrow \infty} a_n = 1 \neq 0.$$

The series becomes $\sum_{n=0}^{\infty} 1$, which clearly diverges.

Case 2: $\alpha \neq 0$ and $-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$

For $\alpha \neq 0$, the sequence $\{\cos(n\alpha)\}$ oscillates. Similar to the sine case:

- If α is a rational multiple of π , say $\alpha = \frac{p\pi}{q}$ where $\gcd(p, q) = 1$, then $\cos(n\alpha)$ is periodic with period dividing $2q$. The sequence repeats its values and does not converge to zero.
- If α is an irrational multiple of π , the sequence $\{\cos(n\alpha)\}$ is equidistributed on $[-1, 1]$, meaning it takes on values throughout this interval infinitely often. Therefore, the limit does not exist.

Step 4: Demonstrating Non-Convergence Using Subsequences

For any $\alpha \in (-\frac{\pi}{2}, \frac{\pi}{2})$ with $\alpha \neq 0$:

- There exist infinitely many values of n such that $n\alpha$ is close to $0 \pmod{2\pi}$, giving $\cos(n\alpha) \approx 1$.
- There exist infinitely many values of n such that $n\alpha$ is close to $\pi \pmod{2\pi}$, giving $\cos(n\alpha) \approx -1$.

Since we can extract subsequences that approach different values (1 and -1), the sequence $\{\cos(n\alpha)\}$ does not converge to any single value, and in particular, does not converge to zero.

Step 5: Special Observation for $\alpha \in (-\frac{\pi}{2}, \frac{\pi}{2})$

Note that for $\alpha = 0$, we have $a_n = 1$ for all n , so the limit is clearly not zero.

For $\alpha \neq 0$ in the given range, $\cos(n\alpha)$ oscillates indefinitely and the limit does not exist (or at minimum does not equal zero).

Step 6: Apply the Divergence Test

Since $\lim_{n \rightarrow \infty} \cos(n\alpha)$ either does not exist or is not equal to zero for $-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$, by the Divergence Test, the series:

$$\sum_{n=0}^{\infty} \cos(n\alpha)$$

diverges.

Final Answer

Conclusion

The series $\sum_{n=0}^{\infty} \cos(n\alpha)$ for $-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ **diverges** because:

$\lim_{n \rightarrow \infty} \cos(n\alpha)$ does not equal zero.
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- If $\alpha = 0$: $\lim_{n \rightarrow \infty} \cos(n\alpha) = 1 \neq 0$.
- If $\alpha \neq 0$: The sequence $\{\cos(n\alpha)\}$ oscillates and does not converge to zero.

In all cases, the necessary condition for convergence is violated, so by the Divergence Test, the series must diverge.